

B. Tech S₁ Examination December 2021

Linear Algebra and Calculus

Part A

1. Find the rank of the matrix $\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & -4 \\ 0 & 4 & 0 \end{bmatrix}$

Soln: $\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & -4 \\ 0 & 4 & 0 \end{bmatrix} \sim \begin{bmatrix} -1 & 0 & -4 \\ 0 & 1 & 0 \\ 0 & 4 & 0 \end{bmatrix} \quad R_1 \leftrightarrow R_2$

$\sim \begin{bmatrix} -1 & 0 & -4 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad R_3 \rightarrow R_3 - 4R_2$

$\therefore \text{Rank} = 2$

2. Find the Eigen values of the matrix $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$

what are the eigen values of A^2 , A^{-1} without using its characteristic equation.

Soln: trace of $A = \lambda_1 + \lambda_2$
 $\therefore \lambda_1 + \lambda_2 = 4 \quad \text{--- (1)}$
 $|A| = \lambda_1 \lambda_2 = 3 \quad \text{--- (2)}$

From eqn (1) and (2) $\lambda_1 = 1, \lambda_2 = 3$

Eigen values of A are $1, 3$

Eigen values of A^2 are $1^2, 3^2$ i.e. $1, 9$

Eigen values of A^{-1} are $1, 1/3$

Part B

11a) Solve the following linear system of equations using Gauss elimination method

$$x + 2y - z = 3, \quad 3x - y + 2z = 1$$

$$2x - 2y + 3z = 2$$

Soln:

$$\tilde{A} = \begin{bmatrix} \textcircled{1} & 2 & -1 & : & 3 \\ 3 & -1 & 2 & : & 1 \\ 2 & -2 & 3 & : & 2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & \textcircled{-7} & 5 & -8 \\ 0 & -6 & 5 & -4 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -8 \\ 0 & 0 & 5/7 & 20/7 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - \frac{6}{7} R_2$$

$$\text{rank } A = \text{rank } \tilde{A} = \text{no. of unknowns} = 3$$

$\therefore$ System of equation have unique solution

By Back substitution, $\frac{5}{7}z = \frac{20}{7} \Rightarrow z = 4$

$$-7y + 5z = -8 \Rightarrow 5z = 7y - 8 \Rightarrow y = 4$$

$$x + 2y - z = 3 \Rightarrow x + 2(4) - 4 = 3 \Rightarrow x = -1$$

$$\therefore \text{Solution is } \underline{x = -1, y = 4, z = 4}$$

11 b) Find the Eigen values and Eigen vectors of $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$

Soln:

Characteristic equation is $\lambda^3 - \text{trace } A \lambda^2 + (A_{11} + A_{22} + A_{33})\lambda - |A| = 0$

$$\lambda^3 - 18\lambda^2 + [5 + 20 + 20]\lambda - 0 = 0$$

Eigen values of are $\lambda = 0, 3, 15$

Eigen vector corresponding to $\lambda = 0$

Consider $(A - \lambda I)x = 0$

$$A - 0I = A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

$$\sim \begin{bmatrix} 8 & -6 & 2 \\ 0 & \frac{20}{8} & -\frac{10}{4} \\ 0 & -\frac{10}{4} & \frac{5}{2} \end{bmatrix}$$

$$R_2 \rightarrow R_2 \rightarrow R_2 + \frac{6}{8}R_1$$

$$R_3 \rightarrow R_3 - \frac{2}{8}R_1$$

$$\sim \begin{bmatrix} 8 & -6 & 2 \\ 0 & \frac{20}{8} & -\frac{10}{4} \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

By Back substitution $\frac{20}{8}x_2 - \frac{10}{4}x_3 = 0$

$$x_2 = x_3$$

$$\text{Let } x_2 = 2 \Rightarrow x_3 = 2$$

$$8x_1 - 6x_2 + 2x_3 = 0$$

$$8x_1 = 6x_2 - 2x_3 = 6(2) - 2(2) = 8 \Rightarrow x_1 = 1$$

Eigen vector corresponding to $\lambda = 0$

$$X_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Eigen vector corresponding to $\lambda = 3$

Consider $(A - \lambda I)X = 0$

$$A - \lambda I = \begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 5 & -6 & 2 \\ 0 & -16/5 & -8/5 \\ 0 & -8/5 & -4/5 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - \frac{-6}{5} R_1$$

$$R_3 \rightarrow R_3 - \frac{2}{5} R_1$$

$$\sim \begin{bmatrix} 5 & -6 & 2 \\ 0 & -16/5 & -8/5 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - \frac{1}{2} R_2$$

By Back substitution $-\frac{16}{5}x_2 - \frac{8}{5}x_3 = 0$

$$-2x_2 = x_3$$

$$\text{Let } x_2 = 1$$

$$x_3 = -2$$

$$5x_1 - 6x_2 + 2x_3 = 0$$

$$5x_1 = 6x_2 - 2x_3 = 6 \times 1 - 2 \times (-2) = 10$$

$$x_1 = 2$$

Eigen vector corresponding to $\lambda = 3$

$$X_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

Eigen vector corresponding to $\lambda = 15$

Consider $(A - 15I)x = 0$

$$A - 15I = \begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix}$$

$$\sim \begin{bmatrix} -7 & -6 & 2 \\ 0 & -\frac{20}{7} & -\frac{40}{7} \\ 0 & -\frac{40}{7} & -\frac{80}{7} \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - \frac{6}{7}R_1 \\ R_3 \rightarrow R_3 + \frac{2}{7}R_1 \end{array}$$

By Back substitution,

$$-\frac{20}{7}x_2 - \frac{40}{7}x_3 = 0$$

$$-x_2 - 2x_3 = 0 \Rightarrow x_2 = -2x_3$$

$$\text{Let } x_3 = 1$$

$$x_2 = -2$$

$$-7x_1 - 6x_2 + 2x_3 = 0$$

$$7x_1 = -6x_2 + 2x_3 = -6(-2) + 2(1) = 14$$

$$x_1 = 2$$

Eigen vector corresponding to $\lambda = 15$

$$x_3 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

$\therefore$ matrix of transformation that diagonalize the matrix

$$P = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$$

$$\text{Diagonal matrix } D = P^{-1}AP = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{bmatrix}$$

12 a) solve the following linear system of equations
 using Gauss elimination $2x - 2y + 4z = 0$, $-3x + 3y - 6z + 5w = 15$
 $x - y + 2z = 0$

$$\tilde{A} = \begin{bmatrix} 2 & -2 & 4 & 0 & 0 \\ -3 & 3 & -6 & 5 & 15 \\ 1 & -1 & 2 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 2 & -2 & 4 & 0 & 0 \\ 0 & 0 & 0 & 5 & 15 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - \frac{-3}{2} R_1 \\ R_3 \rightarrow R_3 - \frac{1}{2} R_1 \end{array}$$

$\text{Rank } A = \text{Rank } \tilde{A} = 2 < \text{no. of variables}$

$\therefore$ System of equations have infinite no. of

Solutions

By Back substitution,

$$5w = 15 \Rightarrow w = 3$$

$$2x - 2y + 4z = 0$$

$$\text{Let } y = t_1, \quad z = t_2$$

$$2x = 2y - 4z = 2t_1 - 4t_2$$

$$x = t_1 - 2t_2$$

$\therefore$ Solutions are $x = t_1 - 2t_2$, $y = t_1$, $z = t_2$, $w = 3$

12 b) Find the matrix of transformation that diagonalize the matrix $A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$. Also write the diagonal matrix.

Soln:

Characteristic equation is $\lambda^2 - \text{trace } A \lambda + |A| = 0$

$$\lambda^2 - 7\lambda + 6 = 0$$

Eigen values are $\lambda = -1, -6$

Eigen vector corresponding to $\lambda = -1$

Consider $(A - \lambda I)x = 0$

$$A + I = \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix}$$

$$\sim \begin{bmatrix} -4 & 2 \\ 0 & 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - \frac{2}{-4} R_1$$

By Back substitution, $-4x_1 + 2x_2 = 0$

$$4x_1 = 2x_2 \Rightarrow 2x_1 = x_2$$

$$\text{Let } x_1 = 1 \Rightarrow x_2 = 2$$

Eigen vector corresponding to $\lambda = -1$

$$X_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Eigen vector corresponding to $\lambda = -6$

Consider $(A - \lambda I)X = 0$

$$A + 6I = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad R_2 \rightarrow R_2 - 2R_1$$
$$\sim \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

By Back substitution,

$$x_1 + 2x_2 = 0 \Rightarrow x_1 = -2x_2$$

$$\text{Let } x_2 = 1 \quad x_1 = -2$$

Eigen vector corresponding to $\lambda = -1$

$$X_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

The matrix of transformation that diagonalize the matrix

$$P = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$$

$$D = P^{-1}AP = \begin{bmatrix} -1 & 0 \\ 0 & -6 \end{bmatrix}$$

③ If $z = \frac{xy}{x^2+y^2}$, find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$

$$z_x = \frac{y^3 - x^2y}{(x^2+y^2)^2}$$

Module 2

$$\frac{\partial z}{\partial y} = \frac{x^3 - y^2x}{(x^2+y^2)^2}$$

4. Show that the eqn $u(x,t) = \sin(x-ct)$ satisfies wave eqn $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$

$$\frac{\partial^2 u}{\partial t^2} = u_{tt}$$

$$u_t = \cos(x-ct) \times -c$$

$$\begin{aligned} u_{tt} &= -c^2 \times -\sin(x-ct) \\ &= c^2 \sin(x-ct) \longrightarrow (1) \end{aligned}$$

$$u_x = \cos(x-ct) \times 1$$

$$u_{xx} = -\sin(x-ct) \longrightarrow (2)$$

From eqn (1) and (2)

$$\underline{\underline{\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}}}$$

13. a) The length and width of a rectangle are measured with errors of at most 3% and 4% respectively. Use differentials to approximate the max. % error in the calculated area.

b) Find the local linear approximation L of $(x,y,z) = xyz$ at the point $P(1,2,3)$. Compute the error in approximation f by L

at the point $Q(1.001, 2.002, 3.003)$.

a) Area

$$A = xy$$

$$A_0 = x_0 y_0$$

$$dA = A_x(x_0, y_0) dx + A_y(x_0, y_0) dy$$

$$A = xy$$

$$A_x = y, \quad A_y = x$$

$$A_x(x_0, y_0) = y_0$$

$$A_y(x_0, y_0) = x_0$$

$$dA = y_0 dx + x_0 dy$$

$$\text{If } dx = \Delta x \text{ and } dy = \Delta y$$

$$\Delta A \approx dA$$

$$\Delta x = x - x_0$$

$$\Delta y = y - y_0$$

$$\left| \frac{\Delta x}{x_0} \right| \leq 0.03$$

$$\left| \frac{\Delta y}{y_0} \right| \leq 0.04$$

$$\frac{\Delta A}{A_0} = \frac{y_0 \Delta x + x_0 \Delta y}{x_0 y_0}$$

$$= \frac{\Delta x}{x_0} + \frac{\Delta y}{y_0}$$

$$\left| \frac{dA}{A_0} \right| \leq \left| \frac{\Delta x}{x_0} \right| + \left| \frac{\Delta y}{y_0} \right|$$

$$= 0.03 + 0.04$$

$$= 0.07$$

$$= 7\%$$

$\therefore$ The max. % error in area

$$= 7\%$$

$$b) \quad L(x, y, z) = f(x_0, y_0, z_0) + f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0)$$

$$(x_0, y_0, z_0) = 1, 2, 3$$

$$f(1, 2, 3) = 6$$

$$f_x = yz = 6$$

$$f_y = xz = 3$$

$$f_z = xy = 2$$

$$L(x, y, z) = 6 + 6(1.001 - 1) + 3(2.002 - 2) + 2(3.003 - 3) \\ = 6.018$$

$$\text{Error} = 0.018$$

distance btw ~~the~~ p and Q

$$= \sqrt{(1.001-1)^2 + (2.002-2)^2 + (3.003-3)^2}$$

$$= \sqrt{0.01}$$

$\therefore$ Error is less than distance btw the points

14 a) Let f be a differentiable function of three variables and suppose that $w = f(x-y, y-z, z-x)$, show that $\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} +$

$$\frac{\partial w}{\partial z} = 0$$

b) Locate all relative extrema of $f(x, y) =$
 $f(x, y) = 4xy - y^4 - x^4$

a) Let $(x-y) = a$, $(y-z) = b$, $(z-x) = c$

$$\frac{\partial \omega}{\partial x} = \frac{\partial \omega}{\partial a} \times \frac{\partial a}{\partial x} + \frac{\partial \omega}{\partial c} \cdot \frac{\partial c}{\partial x}$$

$$= \frac{\partial \omega}{\partial a} \times 1 + \frac{\partial \omega}{\partial c} \times -1$$

$$= \frac{\partial \omega}{\partial a} - \frac{\partial \omega}{\partial c} \longrightarrow \textcircled{1}$$

$$\frac{\partial \omega}{\partial y} = \frac{\partial \omega}{\partial a} \times \frac{\partial a}{\partial y} + \frac{\partial \omega}{\partial b} \times \frac{\partial b}{\partial y}$$

$$= \frac{\partial \omega}{\partial a} \times -1 + \frac{\partial \omega}{\partial b} \times 1$$

$$= -\frac{\partial \omega}{\partial a} + \frac{\partial \omega}{\partial b} \longrightarrow \textcircled{2}$$

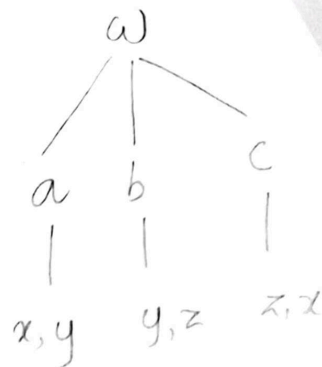
$$\frac{\partial \omega}{\partial z} = \frac{\partial \omega}{\partial b} \times \frac{\partial b}{\partial z} + \frac{\partial \omega}{\partial c} \times \frac{\partial c}{\partial z}$$

$$= \frac{\partial \omega}{\partial b} \times -1 + \frac{\partial \omega}{\partial c} \times 1$$

$$= -\frac{\partial \omega}{\partial b} + \frac{\partial \omega}{\partial c} \longrightarrow \textcircled{3}$$

From eqn $\textcircled{1}$, $\textcircled{2}$ and $\textcircled{3}$

$$\frac{\partial \omega}{\partial x} + \frac{\partial \omega}{\partial y} + \frac{\partial \omega}{\partial z} = 0$$



$$b) f(x, y) = 4xy - y^4 - x^4$$

$$f_x = 4y - 4x^3$$

$$f_y = 4x - 4y^3$$

Critical point is obtained by solving

$$f_x = 0 \text{ and } f_y = 0$$

$$f_x = 0$$

$$\Rightarrow 4y - 4x^3 = 0$$

$$y = x^3 \longrightarrow (1)$$

$$f_y = 0$$

$$\Rightarrow 4x - 4y^3 = 0$$

$$x = y^3 \longrightarrow (2)$$

Substitute eqn (1) in (2)

$$x = (x^3)^3$$

$$x = x^9$$

$$x - x^9 = 0$$

$$x(1 - x^8) = 0$$

$$x = 0$$

$$1 - x^8 = 0$$

$$x^8 = 1 \Rightarrow x = 1 \text{ or } -1$$

$$y = x^3$$

$$\text{If } x = 0, y = 0$$

$$x = 1, y = 1$$

$$x = -1, y = -1$$

$\therefore$ The critical points are

$$(0, 0) \quad (1, 1) \quad (-1, -1)$$

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

$$f_{xx} = -12x^2$$

$$f_{yy} = -12y^2$$

$$f_{xy} = 4$$

at (0, 0)

$$D = (-12 \times 0) (-12 \times 0) - 4^2$$

$$= -16$$

$$D < 0$$

$\therefore (0, 0)$ is a saddle point.

at (1,1)

$$D = (-12 \times 1^2) \times (-12 \times 1^2) - 4^2$$

$$= (-12 \times -12) - 4^2$$

$$= 144 - 16$$

$$= 128$$

$$D > 0$$

$$\therefore f_x = -12$$

$$f_x < 0$$

$\therefore$ (Ans)

f has relative maximum at (1,1)

at (-1,-1)

$$D = -12 \times (-1)^2 \times -12 \times (-1)^2 - 4^2$$

$$= 144 - 16$$

$$= 128$$

$$D > 0$$

$$f_x = -12, f_x \neq 0$$

$\therefore f$ has relative maximum at (-1,-1)

Module - 3

15. a) Find the area bounded by the parabolas $y^2 = 4x$ and $x^2 = \frac{y}{2}$

b) Evaluate $\int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} e^{-(x^2+y^2)} dx dy$ by using polar co-ordinates.

ans: a) The limits of y are,

$$y = 2x^2, \quad y = 2\sqrt{x}$$

$$\text{Given } x^2 = \frac{y}{2}, \text{ and } y^2 = 4x$$

$$\Rightarrow \frac{y^2}{4} = x$$

$$\frac{y^4}{16} = \frac{y}{2}$$

$$\Rightarrow y^3 = 8 \Rightarrow y = 2$$

$$x^2 = \frac{y}{2} \Rightarrow x^2 = 1 \Rightarrow x = 1$$

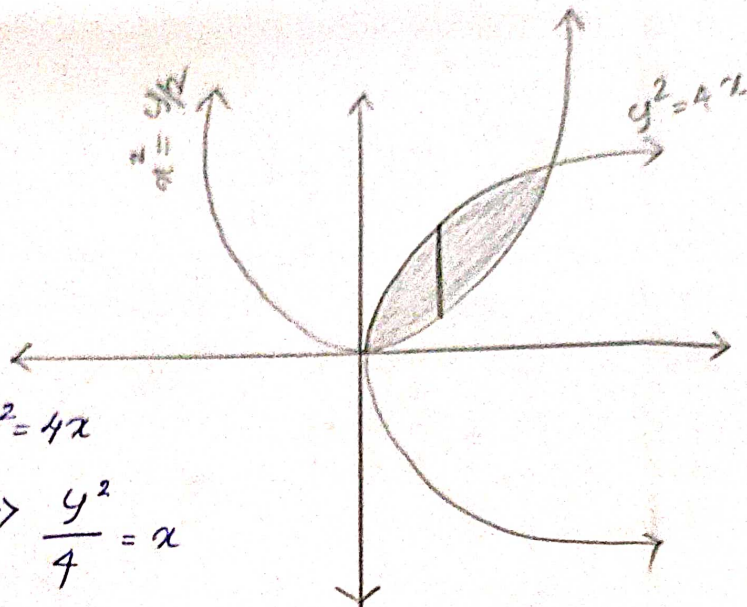
$\therefore$ limits of x are $x=0$ and $x=1$.

$$\text{Area} = \iint_R dA = \int_0^1 \int_{2x^2}^{2\sqrt{x}} dy dx = \int_0^1 [y]_{2x^2}^{2\sqrt{x}} dx$$

$$= \int_0^1 (2\sqrt{x} - 2x^2) dx$$

$$= 2 \left[\frac{2}{3} [x^{3/2}]_0^1 - \frac{2}{3} [x^3]_0^1 \right]$$

$$= 2 \left[\frac{2}{3} - \frac{1}{3} \right] = \underline{\underline{\frac{2}{3}}}$$



b) To change into polar co-ordinates ,

$$x = r \cos \theta, \quad y = r \sin \theta, \quad x^2 + y^2 = r^2, \quad dy dx = r dr d\theta.$$

$$\int_0^{2\pi} \int_0^2 e^{-x^2} x dr d\theta$$

$$= \int_0^{2\pi} \int_0^2 \frac{e^{-t}}{2} dt d\theta$$

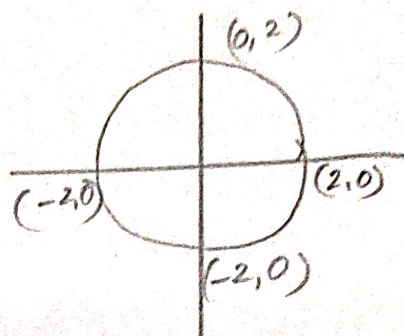
$$= \frac{1}{2} \int_0^{2\pi} \left[\frac{e^{-t}}{-1} \right]_0^2 d\theta$$

$$= \frac{-1}{2} \int_0^{2\pi} \left[e^{-x^2} \right]_0^2 d\theta = \frac{-1}{2} \int_0^{2\pi} (e^{-4} - e^0) d\theta$$

$$= -\frac{1}{2} \int_0^{2\pi} (e^{-4} - 1) d\theta$$

$$= -\frac{1}{2} (e^{-4} - 1) [\theta]_0^{2\pi}$$

$$= \underline{\underline{\pi (1 - e^{-4})}}$$



16. a) Evaluate $\int_0^1 \int_y^1 \frac{x}{x^2 + y^2} dx dy$ by reversing the order of integration

b) Use triple integral to find the volume of the solid within the cylinder $x^2 + y^2 = 9$ and between the planes $z = 1$ and $x + z = 5$.

b)

ans:- a)

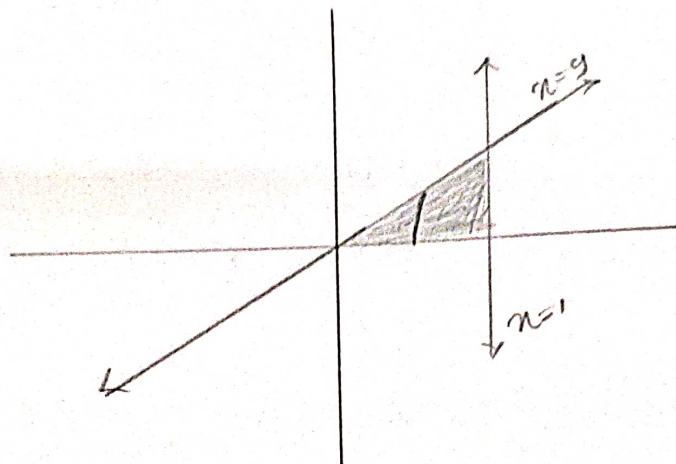
$$\int_0^1 \int_y^1 \frac{x}{x^2+y^2} dx dy$$

$$= \int_0^1 \int_0^x \frac{x}{x^2+y^2} dy dx$$

$$= \int_0^1 x \left[\frac{1}{x} \tan^{-1} \frac{y}{x} \right]_0^x dx$$

$$= \int_0^1 \frac{x}{x} [\tan^{-1} 1 - \tan^{-1} 0] dx$$

$$= \int_0^1 \frac{\pi}{4} dx = \frac{\pi}{4} [x]_0^1 = \underline{\underline{\frac{\pi}{4}}}$$

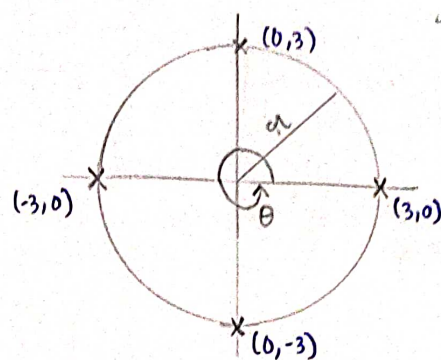


$$b) z \rightarrow 1 \text{ to } 5-x$$

$$x^2 + y^2 = 9$$

x varies from 0 to 3

θ varies from 0 to 2π



Using triple integrals

$$\int_0^{2\pi} \int_0^3 \int_1^{5-x} dz \, r \, dr \, d\theta$$

$$\Rightarrow \int_0^{2\pi} \int_0^3 [z]_1^{5-x} r \, dr \, d\theta \Rightarrow \int_0^{2\pi} \int_0^3 (5-x-1) r \, dr \, d\theta$$

$$\Rightarrow \int_0^{2\pi} \int_0^3 (5-r\cos\theta-1) r \, dr \, d\theta \Rightarrow \int_0^{2\pi} \int_0^3 (5r - r^2\cos\theta - r) \, dr \, d\theta$$

$$\Rightarrow \int_0^{2\pi} \left[5 \frac{r^2}{2} - \cos\theta \frac{r^3}{3} - \frac{r^2}{2} \right]_0^3 d\theta \Rightarrow \int_0^{2\pi} \left(\frac{45}{2} - 9\cos\theta - \frac{9}{2} \right) d\theta$$

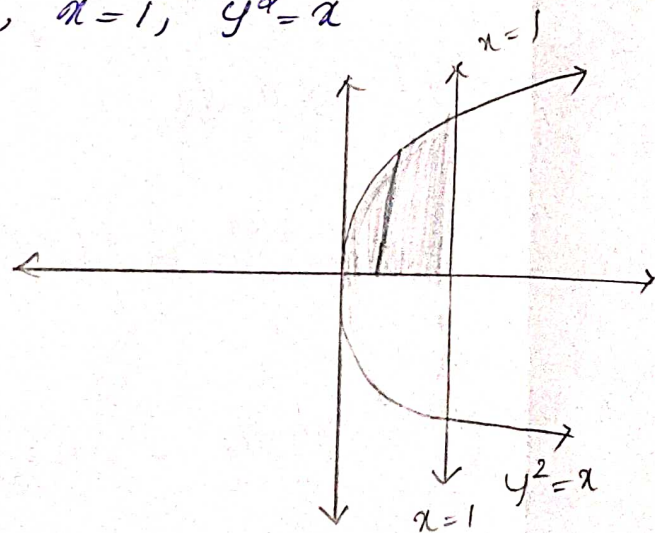
$$\Rightarrow \int_0^{2\pi} 18 - 9\cos\theta \, d\theta \Rightarrow \int_0^{2\pi} 18 \, d\theta - \int_0^{2\pi} 9\cos\theta \, d\theta$$

$$\Rightarrow \left[18 [\theta]_0^{2\pi} - 9 [\sin\theta]_0^{2\pi} \right] \Rightarrow \underline{36\pi}$$

$$\begin{aligned}
 5. \int_0^3 \int_0^2 \int_0^1 xyz \, dx \, dy \, dz &= \int_0^3 \int_0^2 yz \left[\frac{x^2}{2} \right]_0^1 dy \, dz \\
 &= \frac{1}{2} \int_0^3 z \cdot \left[\frac{y^2}{2} \right]_0^2 dz \\
 &= \frac{1}{2} \times \frac{4}{2} \int_0^3 z \, dz \\
 &= \frac{1}{2} \times 2 \times \left[\frac{z^2}{2} \right]_0^3 = 1 \times \frac{9}{2} \\
 &= \underline{\underline{\frac{9}{2}}}
 \end{aligned}$$

6. Find the mass of the lamina with density $\delta(x,y) = x+2y$ is bounded by the x -axis, $x=1$, $y^2=x$

ans:
$$\begin{aligned}
 &\int_0^1 \int_0^{\sqrt{x}} (x+2y) \, dy \, dx \\
 &= \int_0^1 \left[x[y]_0^{\sqrt{x}} + [y^2]_0^{\sqrt{x}} \right] dx \\
 &= \int_0^1 (x^{3/2} + x) \, dx
 \end{aligned}$$



$$\begin{aligned}
 &= \frac{2}{5} \left[x^{5/2} \right]_0^1 + \left[\frac{x^2}{2} \right]_0^1 = \frac{2}{5} + \frac{1}{2} \\
 &= \underline{\underline{\frac{9}{10}}}
 \end{aligned}$$

PART-A

7. Find the rational number represented by the repeating decimal $5.373737 - \dots$

Ans: $5.373737 - \dots = 5 + .37 + .0037 + .00037 + \dots$

$$.37 + .0037 + .00037 + \dots = \frac{37}{99}$$

$$\therefore 5.373737 - \dots = 5 + \frac{37}{99} = \frac{532}{99}$$

8. Examine the convergence of $\sum_{k=1}^{\infty} \frac{k^2}{2k^2+3}$.

Ans: Consider $\sum \frac{k^2}{k^2} = \sum 1$ which is divergent.

$$\begin{aligned} \lim_{k \rightarrow \infty} \frac{a_k}{b_k} &= \lim_{k \rightarrow \infty} \frac{k^2}{2k^2+3} = \lim_{k \rightarrow \infty} \frac{k^2}{k^2(2+\frac{3}{k^2})} \\ &= 1 > 0 \text{ \& finite.} \end{aligned}$$

by limit comparison test, since $\sum 1$ is divergent

$$\sum_{k=1}^{\infty} \frac{k^2}{2k^2+3} \text{ is also divergent.}$$

Module - 4

17a) Test the convergence of (i) $\sum_{k=1}^{\infty} \frac{3k^3 - 2k^2 + 4}{k^7 - k^3 + 2}$

(ii) $\sum_{k=1}^{\infty} \frac{k^k}{k!}$

Sol: i) Let $a_k = \frac{3k^3 - 2k^2 + 4}{k^7 - k^3 + 2}$

$$b_k = \frac{3}{k^4}$$

$$\begin{aligned} \lim_{k \rightarrow \infty} \frac{a_k}{b_k} &= \lim_{k \rightarrow \infty} \frac{3k^3 - 2k^2 + 4}{k^7 - k^3 + 2} \cdot \frac{k^4}{3} \\ &= \lim_{k \rightarrow \infty} \frac{3k^7 - 2k^6 + 4k^4}{3(k^7 - k^3 + 2)} \\ &= \lim_{k \rightarrow \infty} \frac{3k^7 \left(1 - \frac{2k^6}{3k^7} + \frac{4k^4}{3k^7}\right)}{3k^7 \left(1 - \frac{k^3}{k^7} + \frac{2}{k^7}\right)} = 1 \end{aligned}$$

By limit comparison test $\sum a_k$ converges.
Since $\sum b_k$ is a convergent series by p-test.

ii) $a_k = \frac{k^k}{k!}$

$$a_{k+1} = \frac{(k+1)^{k+1}}{(k+1)!}$$

$$\begin{aligned} \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} &= \lim_{k \rightarrow \infty} \frac{(k+1)^{k+1}}{(k+1)!} \cdot \frac{k!}{k^k} = \lim_{k \rightarrow \infty} \frac{(k+1)^k \cdot (k+1)}{k^k \cdot (k+1)} \\ &= \lim_{k \rightarrow \infty} \left(\frac{k+1}{k}\right)^k = \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k}\right)^k = e > 1 \end{aligned}$$

By ratio test, the series diverges.

17 b) Test whether the series $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{1}{\sqrt{k+1}}$ is absolutely convergent or conditionally convergent.

Ans:

$$\text{Let } a_k = \frac{1}{\sqrt{k+1}}$$

$$\lim_{k \rightarrow \infty} a_k = \lim_{k \rightarrow \infty} \frac{1}{\sqrt{k+1}} = 0$$

$$\frac{a_{k+1}}{a_k} = \frac{1}{\sqrt{k+1}+1} \cdot \sqrt{k+1} < 1$$

$\therefore$ by Leibnitz test, the series converges.

$$u_k = (-1)^{k+1} \frac{1}{\sqrt{k+1}}$$

$$|u_k| = \frac{1}{\sqrt{k+1}}$$

$$\therefore v_k = \frac{1}{\sqrt{k}}$$

$$|u_k| < v_k$$

by limit comparison test, $\sum |u_k|$ diverges
 $\Rightarrow \sum u_k$ diverges absolutely.

$\Rightarrow$ conditionally convergent.

18.a) Test the convergence of the series

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \frac{1}{4 \cdot 5} + \dots$$

Ans:

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots = \sum_{k=1}^{\infty} \frac{1}{k(k+1)}$$

$$\text{Let } a_k = \frac{1}{k(k+1)}, \quad b_k = \frac{1}{k^2}$$

$$\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \lim_{k \rightarrow \infty} \frac{k^2}{k^2+1} = 1$$

$\therefore$ by limit comparison test, $\sum a_k$ converges
since $\sum b_k$ is a convergent p-series.

(86) Test the convergence of i) $\sum_{k=1}^{\infty} \left(\frac{k}{k+1}\right)^{k^2}$ ii) $\sum_{k=1}^{\infty} \frac{7^k}{k!}$

Ans: i) Let $a_k = \left(\frac{k}{k+1}\right)^{k^2}$

$$\begin{aligned} \text{by root test } \lim_{k \rightarrow \infty} (a_k)^{1/k} &= \lim_{k \rightarrow \infty} \left[\left(\frac{k}{k+1}\right)^{k^2} \right]^{1/k} \\ &= \lim_{k \rightarrow \infty} \left(\frac{k}{k+1}\right)^k = \lim_{k \rightarrow \infty} \frac{k^k}{(k+1)^k} \\ &= \lim_{k \rightarrow \infty} \frac{k^k}{k^k \left(1 + \frac{1}{k}\right)^k} = \frac{1}{e} < 1 \end{aligned}$$

$\therefore$ the series converges.

$$\text{ii) } a_k = \frac{7^k}{k!}$$

$$a_{k+1} = \frac{7^{k+1}}{(k+1)!}$$

$$\text{by ratio test, } \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = \lim_{k \rightarrow \infty} \frac{(k+1)!}{k!} \cdot \frac{7^k}{7^{k+1}}$$

$$= \lim_{k \rightarrow \infty} \frac{7^{k+1}}{(k+1)!} \cdot \frac{k!}{7^k} \quad \text{[cancel terms]}$$

$$= \lim_{k \rightarrow \infty} \frac{7}{k+1} = 0 < 1$$

$\therefore \sum a_k$ converges.

PART A

9. Find the Taylor series expansion of $f(x) = \sin \pi x$ about $x = \frac{1}{2}$

Soln: Taylor series expansion of $f(x)$ about $x = x_0$ is $f(x) = f(x_0) + \frac{x-x_0}{1!} f'(x_0) + \frac{(x-x_0)^2}{2!} f''(x_0) + \dots$

$$f(x) = \sin \pi x$$

$$f(1/2) = \sin \pi/2 = 1$$

$$f'(x) = \cos \pi x \times \pi$$

$$f'(1/2) = \pi \times \cos \pi/2 = 0$$

$$f''(x) = \pi \times \sin \pi x \times \pi = -\pi^2 \sin \pi x$$

$$f''(1/2) = -\pi^2 \sin \pi/2 = -\pi^2$$

$$f'''(x) = -\pi^3 \cos \pi x$$

$$f'''(1/2) = -\pi^3 \cos \pi/2 = 0$$

$$f^{IV}(x) = +\pi^4 \sin \pi x$$

$$f^{IV}(1/2) = \pi^4 \sin \pi/2 = \pi^4$$

$$\therefore f(x) = 1 + \frac{x-1/2}{1!} \times 0 + \frac{(x-1/2)^2}{2!} \times -\pi^2 + \frac{(x-1/2)^3}{3!} \times 0 + \frac{(x-1/2)^4}{4!} \times \pi^4 + \dots$$

$$f(x) = 1 + x^2 \frac{(x-1/2)^2}{2!} + x^4 \frac{(x-1/2)^4}{4!} - \dots$$

10. If $f(x)$ is a periodic function with period 2π defined in $[-\pi, \pi]$. Write the Euler's formula a_0, a_n, b_n for $f(x)$.

$$a_0 = \frac{1}{2L} \int_{\alpha}^{\alpha+2L} f(x) dx$$

$$\therefore a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{L} \int_{\alpha}^{\alpha+2L} f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$\therefore a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

$$b_n = \frac{1}{L} \int_{\alpha}^{\alpha+2L} f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$\therefore b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx$$

19 a) Find the Fourier series expansion of

$$f(x) = x - x^2 \text{ in the range } (-1, 1).$$

Soln: Fourier series expansion of $f(x)$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{1}{2l} \int_a^{a+2l} f(x) dx$$

$$= \frac{1}{2} \int_{-1}^1 x - x^2 dx$$

$$= \frac{1}{2} \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_{-1}^1$$

$$= \frac{1}{2} \left[\frac{1}{2} - \frac{1}{3} - \left(\frac{1}{2} - \frac{-1}{3} \right) \right]$$

$$= \frac{1}{2} \times \frac{-2}{3} = \underline{\underline{\frac{-1}{3}}}$$

$$a_n = \frac{1}{l} \int_a^{a+2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \int_{-1}^1 (x - x^2) \cos(n\pi x) dx$$

$$= (x - x^2) \frac{\sin(n\pi x)}{n\pi} - \left[(1 - 2x) \cdot \frac{-\cos(n\pi x)}{n^2 \pi^2} \right] + (-2) \cdot \left[\frac{-\sin(n\pi x)}{n^3 \pi^3} \right] \Bigg|_{-1}^1$$

$$= \left[(1-1) \sin\left(\frac{n\pi x}{n\pi}\right) - \left(-\frac{1-1}{n\pi}\right) \sin(-n\pi) + \frac{1}{n^2\pi^2} ((1-2) \cos n\pi) \right. \\ \left. - (1+2) \cos(-n\pi) + \frac{2}{n^3\pi^3} (\sin n\pi - \sin(-n\pi)) \right]$$

$$= 0 - \frac{4}{n^2\pi^2} \cos n\pi$$

$$= \frac{-4}{n^2\pi^2} (-1)^n$$

$$b_m = \frac{1}{l} \int_{-l}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \int_{-1}^1 (x-x^2) \sin(n\pi x) dx$$

$$= (x-x^2) \cdot \frac{-\cos n\pi x}{n\pi} - (1-2x) x \cdot \frac{-\sin(n\pi x)}{n^2\pi^2} \\ + (-2) \cdot \frac{\cos(n\pi x)}{n^3\pi^3} \Big]_{-1}^1$$

$$= \frac{-1}{n\pi} [0 - -2x \cos n\pi] - 0 - \frac{2}{n^3\pi^3} [\cos n\pi - \cos n\pi]$$

$$= \frac{-2}{n\pi} \cos n\pi$$

$$= \frac{-2}{n\pi} (-1)^n$$

$$\therefore f(x) = -\frac{1}{3} + \sum_{n=1}^{\infty} \frac{-4}{n^2\pi^2} (-1)^n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} \frac{-2}{n\pi} (-1)^n \sin\left(\frac{n\pi x}{l}\right)$$

b) Obtain the half range Fourier cosine series of $f(x) = e^{-x}$ in $0 < x < 2$.

Soln Half range Fourier cosine series of

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{1}{l} \int_0^l f(x) dx$$

$$= \frac{1}{2} \int_0^2 e^{-x} dx = \frac{1}{2} \left[\frac{e^{-x}}{-1} \right]_0^2 = -\frac{1}{2} [e^{-2} - e^0]$$

$$= -\frac{1}{2} [e^{-2} - 1]$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{2} \int_0^2 e^{-x} \cos\left(\frac{n\pi x}{2}\right) dx$$

$$= \frac{e^{-x}}{\left(\frac{1}{2}\right)^2 + \frac{n^2\pi^2}{4}} \left[\frac{1}{2} \cos\left(\frac{n\pi x}{2}\right) + \frac{n\pi}{2} \sin\left(\frac{n\pi x}{2}\right) \right]_0^2$$

$$= \frac{e^{-2}}{1 + \frac{n^2\pi^2}{4}} \left[-\cos(n\pi) + \frac{n\pi}{2} \times \sin(n\pi) \right] - \left[\left(\frac{e^{-0}}{1 + \frac{n^2\pi^2}{4}} \right) (\cos 0 + 0) \right]$$

$$= \frac{e^{-2}}{1 + \frac{n^2\pi^2}{4}} \left[(-1)^n \right] + \left[\frac{1}{1 + \frac{n^2\pi^2}{4}} \right]$$

$$= \frac{1 - e^{-2}(-1)^n}{1 + \frac{n^2\pi^2}{4}}$$

$$\therefore f(x) = -\frac{1}{2} [e^{-2} - 1] + \sum_{n=1}^{\infty} \frac{1 - e^{-2}(-1)^n}{1 + \frac{n^2\pi^2}{4}} \cos\left(\frac{n\pi x}{2}\right)$$

so a) Find the fourier series expansion of $f(x) = x^2$ in the interval $-\pi < x < \pi$. Hence show that $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$

Soln: Since $f(x) = x^2$ is an even function, $b_n = 0$.

$$a_0 = \frac{1}{2l} \int_{-l}^l f(x) dx$$

$$= \frac{2}{2l} \int_0^l f(x) dx \quad (\because f(x) \text{ is even})$$

$$= \frac{1}{l} \int_0^l f(x) dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x^2 dx = \frac{1}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi} = \frac{1}{\pi} \left[\frac{\pi^3}{3} \right]$$

$$= \frac{\pi^2}{3}$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{l} \int_0^l x^2 \cos\left(\frac{n\pi x}{\pi}\right) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x^2 \cos(nx) dx$$

$$= \frac{2}{\pi} \left[x^2 \times \frac{\sin(nx)}{n} - 2x \times \frac{\cos(nx)}{n^2} + 2x \times \frac{\sin nx}{n^3} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left\{ \left[\pi^2 \cdot \frac{\sin n\pi}{n} - 0 \right] + \frac{2\pi}{n^2} (\cos n\pi - 1) - \frac{2}{n^3} (\sin n\pi - \sin 0) \right\}$$

$$= \frac{2}{\pi} \left[0 + \frac{2\pi}{n^2} (-1)^n - 0 \right]$$

$$= \frac{4\pi}{\pi n^2} (-1)^n$$

$$= \frac{4}{n^2} (-1)^n$$

$$\therefore f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^n \cos nx$$

$$\text{Put } x = \frac{\pi^2}{3} \Rightarrow \frac{\pi^2}{3} = \frac{\pi^2}{3} + 4 \left[-\frac{\cos 2x}{1^2} + \frac{\cos 2x}{2^2} - \frac{\cos 3x}{3^2} + \dots \right]$$

$$\frac{\pi^2}{3} = \frac{\pi^2}{3} + 4 \left[\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots \right]$$

$$= 4 \left[\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots \right]$$

$$= 1 - \frac{1}{2^2} + \frac{1}{3^2} - \dots$$

20 b) Obtain the half range Fourier sine series

$$\text{of } f(x) = \begin{cases} x, & 0 < x < \pi/2 \\ \pi - x, & \pi/2 < x < \pi \end{cases}$$

Soln: Fourier sine series for $f(x)$ is,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \left(\frac{n\pi x}{l} \right)$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(nx) dx$$

$$= \frac{2}{\pi} \left[\int_0^{\pi/2} x \sin(nx) dx + \int_{\pi/2}^{\pi} (\pi-x) \sin(nx) dx \right]$$

$$= \frac{2}{\pi} \left\{ x \cdot \frac{-\cos nx}{n} - \left[\frac{-\sin nx}{n^2} \right]_0^{\pi/2} + (\pi-x) \cdot \frac{-\cos nx}{n} - (-1) \cdot \frac{-\sin nx}{n^2} \right\}_{\pi/2}^{\pi}$$

$$= \frac{2}{\pi} \left\{ \left[\frac{-\frac{\pi}{2} \cos(\frac{n\pi}{2})}{n} + \frac{1}{n^2} \sin(\frac{n\pi}{2}) \right] - \left[0 - 0 - \frac{\pi}{2} \cdot \frac{\cos \frac{n\pi}{2}}{n} - \left(\frac{\sin \frac{n\pi}{2}}{n^2} \right) \right] \right\}$$

$$= \frac{2}{\pi} \cdot \frac{2}{n^2} \sin \frac{n\pi}{2}$$

$$= \frac{4}{\pi n^2} \sin \frac{n\pi}{2}$$

$$\therefore f(x) = \sum_{n=1}^{\infty} \frac{4}{n^2 \pi} \sin \frac{n\pi}{2} \sin nx$$

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